

Unit 2 Practice Problems

Transformations Practice

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 2 Practice Problems

Directions

Work each problem without looking at the answer bank. These Unit 2 problems include an answer bank only. Detailed worked solutions begin in Unit 3 as requested.

Problems

1. Translate $(-2, 5)$ by $(x, y) \mapsto (x + 4, y - 3)$.
2. Reflect $(3, -4)$ over the x -axis.
3. Reflect $(-7, 2)$ over the y -axis.
4. Reflect $(5, -1)$ over the line $y = x$.
5. Rotate $(6, -2)$ 90° counterclockwise about the origin.
6. Rotate $(-3, 8)$ 180° about the origin.
7. Rotate $(4, 9)$ 270° counterclockwise about the origin.
8. Triangle ABC has $A(1, 1)$, $B(4, 1)$, $C(2, 5)$. Translate by $\langle -5, 2 \rangle$. Give A' , B' , and C' .
9. Start with $(3, 4)$. Reflect over the y -axis, then translate by $\langle 2, -1 \rangle$.
10. Is a translation a rigid motion? State what it preserves.
11. Is a dilation with scale factor 3 a rigid motion?
12. A segment is reflected and then rotated. What happens to its length?
13. Reflect the line $x = 2$ over the y -axis.
14. Reflect the line $y = -3$ over the x -axis.
15. How many reflection symmetries and what rotational symmetry order does a non-square rectangle have?
16. How many reflection symmetries and what rotational symmetry order does a square have?
17. What is the smallest positive rotation that maps a regular pentagon onto itself?
18. When two parallel lines are cut by a transversal, what is true of corresponding angles?
19. When two parallel lines are cut by a transversal, what is true of alternate interior angles?
20. When two parallel lines are cut by a transversal, what is true of same-side interior angles?
21. Alternate interior angles measure $4x + 8$ and $6x - 22$. If the lines are parallel, find x and the angle measure.
22. Same-side interior angles measure $3x + 15$ and $5x + 5$. If the lines are parallel, find x and both measures.
23. State the converse theorem using corresponding angles to prove lines parallel.
24. What transformation is equivalent to reflecting over two parallel lines d units apart?
25. If two figures are congruent, what kind of transformations can map one onto the other?

Answer Bank

1. $(2, 2)$.
2. $(3, 4)$.
3. $(7, 2)$.
4. $(-1, 5)$.
5. $(2, 6)$.
6. $(3, -8)$.
7. $(9, -4)$.
8. $A'(-4, 3)$, $B'(-1, 3)$, $C'(-3, 7)$.
9. $(-1, 3)$.
10. Yes; it preserves length, angle measure, parallelism, and shape.
11. No; it changes lengths unless the scale factor is 1.
12. The length stays the same.
13. $x = -2$.
14. $y = 3$.
15. Two reflection lines; rotational symmetry order 2.
16. Four reflection lines; rotational symmetry order 4.
17. 72° .
18. They are congruent.
19. They are congruent.
20. They are supplementary.
21. $x = 15$; each angle is 68° .
22. $x = 20$; measures 75° and 105° .
23. If corresponding angles are congruent, then the lines are parallel.
24. A translation of length $2d$ perpendicular to the parallel lines.
25. A sequence of rigid motions: translations, rotations, and reflections.